

Application Note: Ethernet Adapter

INTRODUCTION

FreeETarget supports a variety of different communications interfaces:

- USB
- WiFi
- RS485
- Token Ring
- Ethernet

This document describes adding an Ethernet adapter to the target. This allows the user to route all of the targets into a robust network that can be easily accessed anywhere on the range.

REQUIRED EQUIPMENT

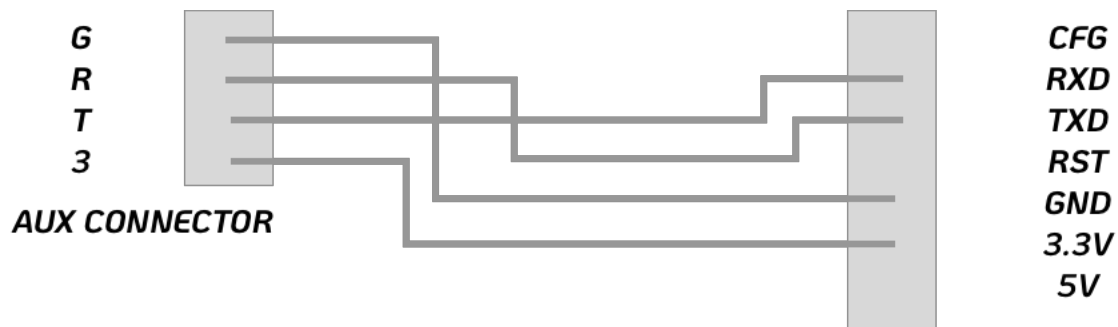
The following items are required.

- FreeETarget Version 5 or higher
- PUSR USR-TCP23-T2 available through a variety of suppliers.
- Cable Harness
- Ethernet cable, CAT 5 or higher



PREPARATION

- Build the cable harness as shown in Figure 1



ESP32

Ethernet

Figure 1: USB-TCP232-T2 Cable Harness

IMPORTANT

When using with Version 5 boards, connect the voltage signal (3 in Figure 1) to the 5VDC pin of the PUSA

- Power on the target.
- Configure the auxiliary interface to Ethernet {"AUX_MODE":16}
- Connect the Ethernet cable to the adapter.
 - Verify that the green LED is illuminated, if not verify the cabling.

CONFIGURATION

The USR-TCP232-T2 can be configured using the built in web browser. Unfortunately, it is very particular about how this interface is accessed. Errors in the configuration may result in bricking the part and may be irretrievably broken with simple tools.

Building the Network

The Ethernet interface is delivered with IP address 192.168.0.7.

Connect the interface to the network, ensuring that there are no other devices on the network with that address. Make sure that your PC is also on a 192.168.0.x network.

Connecting to the Adapter

Open a web browser to 192.168.0.7. You will see the opening screen as shown in Figure 2

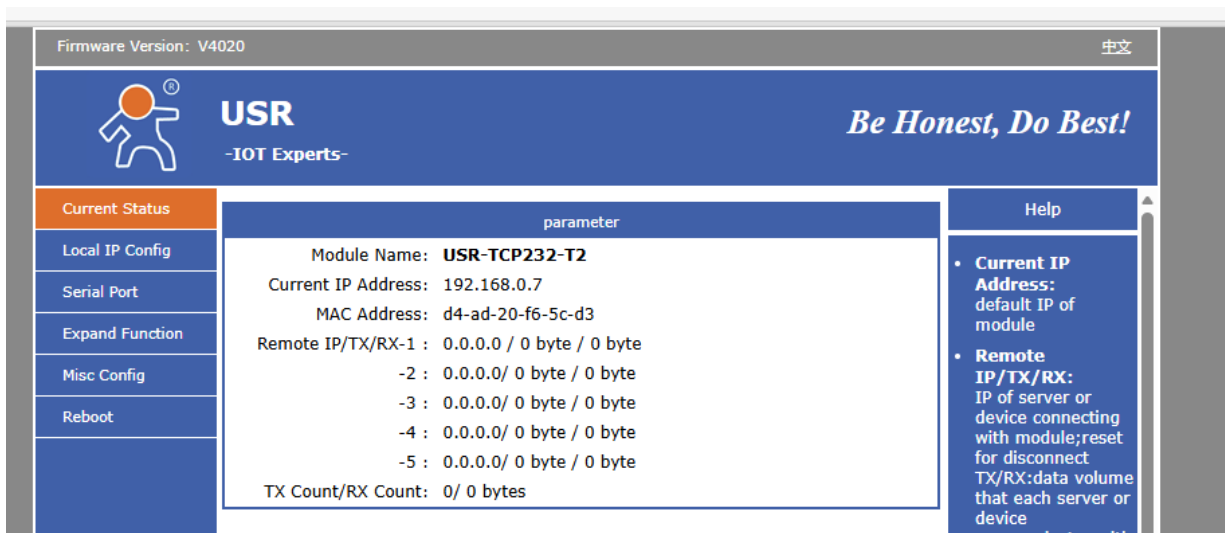


Figure 2: Opening Screen

Setting up the IP address

Manually change the IP address (Static IP) as shown in Figure 3 to the address you want on your network.

The IP address must be unique on that network.

The Gateway and DNS server entries are not used by the target and can be left alone.

Press SAVE

IMPORTANT Record the IP address in the worksheet. If you lose the IP address, there is almost no way to recover the device

Firmware Version: V4020 中文

USR
-IOT Experts-

Be Honest, Do Best!

Current Status

Local IP Config

Serial Port

Expand Function

Misc Config

Reboot

parameter

IP type: Static IP ▼

Static IP: . . .

Submask: . . .

Gateway: . . .

DNS Server: . . .

Help

- **IP type:**
StaticIP or DHCP
- **StaticIP:**
Module's static ip
- **Submask:**
usually
255.255.255.0
- **Gateway:**
Usually router's ip
address
- **DNS IP:**
DNS gateway or
Router's IP

Figure 3: Setting the IP address

Setting up the Interface

Set the Work Mode to Server as shown in Figure 4.

The Local Port Number should be left alone but may be changed to 1090 if desired.

Record the port number in the worksheet.

Press SAVE

Firmware Version: V4020 中文

USR
-IOT Experts-

Be Honest, Do Best!

Current Status

Local IP Config

Serial Port

Expand Function

Misc Config

Reboot

parameter

Baud Rate: bps

Data Size: 8 ▼ bit

Parity: None ▼

Stop Bits: 1 ▼ bit

Local Port Number: (0~65535)

Remote Port Number: (1~65535)

Work Mode: TCP Server ▼

Remote Server Addr:
[192.168.0.201]

RESET: ☐

LINK: ☐

INDEX: ☐

Help

- **UDP multicast:**
In UDP Client mode, the address range of remote server is 224.0.0.2 - 239.255.255.255, which needs to be modified manually
- **HTTPD URL:**
Module add GET/POST and HTTP/1.1 in URL automatically according to user's setting.
- **HTTPD Packet Header:**
Module add HOST

Figure 4: Serial Interface

Miscellaneous Configuration

Do not make any changes to the Miscellaneous Configuration shown in Figure 5

Firmware Version: V4020 中文

USR
-IOT Experts- *Be Honest, Do Best!*

Current Status
Local IP Config
Serial Port
Expand Function
Misc Config
Reboot

parameter

Module Name:

Webserver Port:

Username:

Password:

Max Clients Connect To TCP Server: (1~16)

Reset Timeout: (s)(0,60~65535s)

Help

- **Max Clients Connect To TCP Server:**
when Module is TCP Server, the max number of TCP client allowed to connect
- **Timeout Restart Time:**
When the network port without data, timeout restart, If set to 0s, function

Figure 5: Miscellaneous Configuration

Restart the Adapter

Press the RESTART MODULE to reboot the adapter as shown in Figure 6.

After restarting, the adapter will change to the IP address recorded in the worksheet. The device can be changed by logging into the new address and making changes

Firmware Version: V4020 中文

USR
-IOT Experts- *Be Honest, Do Best!*

Current Status
Local IP Config
Serial Port
Expand Function
Misc Config
Reboot

Reboot

Restart Module

Help

Restart module

Figure 6: Restarting the Module

USING THE ETHERNET ADAPTER

Once the adapter reboots, it will come up as (for example) 192.168.0.7 and can be accessed as if the target was connected to WiFi.

Change the connection type in the FreeETarget configuration as shown in Figure 7

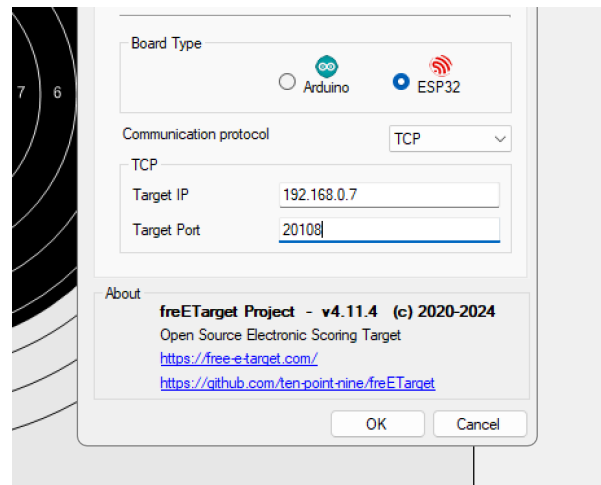


Figure 7: FreeETarget Configuration

The Ethernet Adapter will work with all versions of the PC Client.

Ethernet Work Sheet

[illegible]